



DEALER BEST PRACTICE SERIES

Computerized Torques Wrenches Used to Improve Quality Control

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

Computerized Torques Wrenches Used to Improve
Quality Control0

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1.0 Introduction

Computerized torque wrenches electronically record final torque readings and enable those readings to be stored in a computer file. This information is especially useful when tightening critical bolts using a torque-turn method. This includes cylinder head bolts, connecting rod bolts, counterweight bolts, as well as other applications.

Conventional torque wrenches do not typically measure the final torque after the turn is applied and do not record those values. With a computerized wrench, a technician can easily review the final torque values on all the bolts to ensure the final torques are consistent.

2.0 Best Practice Description

Digital computerized torque wrenches in various sizes are available from a number of different suppliers.

COMPUTERIZED TORQUE WRENCH AND TORQUE ANGLE GAUGE	
	Computerized torque wrench with a range between 60 to 600 lbt-ft used to adjust the main caps, rods and crank counterweight bolts and record the value obtained in an electronic file.
	Torque angle gauge used with the computerized torque wrench to obtain the exact torque angles when adjusting main caps, rods and counterweight bolts.

Example is the CDI Computorq (Snap On) $\frac{3}{4}$ in wrench pictured below.

A recent example of how this tool can improve quality comes from Ferreyros, the Caterpillar dealer in Peru.

When Caterpillar changed the recommended connecting rod bolt lubrication from a 6V4876 lube to Loctite Molypaste 51049, Ferreyros immediately noticed a change in final torque. Using the 6V4876 lube, with 65 ft-lbs followed by a 90 degrees torque turn, the final torque was 150 ft-lbs. With the Molypaste, the final torque was 135 ft-lbs. The Caterpillar recommended lube was changed to a Bel Ray 6770 lubricant as result.

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3.0 Implementation Steps

1. Research market for best digital torque wrench that meet dealer requirements. Purchase wrench
2. Establish standard process to record critical torques and check for bolt consistency.

4.0 Benefits

1. Critical torques are documented and checked for consistency as a quality control process.

5.0 Resources Required

1. Cost of a $\frac{3}{4}$ inch wrench ranges from US \$1,500.00 to \$2,500.00.
2. Establish a file to retain records.

6.0 Supporting Attachments / References

7.0 Related Best Practices

8.0 Acknowledgements

Thanks to Ferreyros for photos and “success story”.

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